2026 IEEE International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE 2026)
29–31 January 2026, Rajshahi-6204, Bangladesh

2026 5th International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE) | 979-8-3315-6135-2/26/$31.00 ©2026 IEEE | DOI: 10.1109/ICECTE69292.2026.11429338

Optimized Transfer Learning Model for Accurate Early Breast Cancer Diagnosis

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***Abstract*—Breast cancer is one of the major causes of death in women worldwide. Although automated machine learning systems have great potential to assist the clinical diagnosis process, their performance is often limited by the lack of data availability and class imbalance of mammographic data. This study proposes a framework of transfer learning based on seven pre-trained convolutional neural networks, such as ResNet50, VGG16, VGG19, InceptionV3, EfficientNetB0, DenseNet121, and MobileNetV2, for binary classification of mammography images. To address the lack of data, geometric augmentation techniques (flipping transformation) were used to expand the data from 745 to 2,980 images. Furthermore, three filtering methods applied in ImageJ, such as 16 color, 6 shade, and spectrum filtering, were tested to improve the visibility of the features. Among the models, DenseNet121 with spectrum filtering performed the best with 98.95% accuracy, 98.55% F1 score, 97.9% Matthews correlation coefficient, and 97.8% Cohen's Kappa were recorded in the ablation studies. Five-fold cross-validation was used to verify the model's robustness, and the accuracy was 98.95% $\pm$ 0.0004%, precision of 98.54%, recall of 98.61%, specificity of 99.23%, and an AUC of 0.99. Overall, spectrum filtering led to about a 6% improvement in performance over traditional preprocessing methods. These results show the potential of transfer learning in the automated identification of breast cancer while showing the importance of preprocessing optimization in medical image classification.**

***Keywords*—Breast cancer detection, transfer learning models, data augmentation, ImageJ filtering, DenseNet121, ablation study**

I. Introduction

Breast cancer is a significant health issue worldwide, with almost 25% of all cancer cases diagnosed in women and over 2.3 million new cases annually [1]. The recent developments in mammographic screening have significantly improved the early detection rates, and deep learning-based systems can decrease the interval cancer rate by as much as 9.4% [2]. Nevertheless, inter-reader agreement in the detection of cancer is still an issue, with an inter-radiologist agreement of 75-85% being noted in the diagnosis of the disease [3].

Despite these improvements, the correct interpretation of mammograms is still challenging because of such factors as the change of the density of the breast that occurs in nearly all women, or subtle morphological differences and distortions in the architecture to determine the benign and malignant lesions [4]. Moreover, existing computer-aided detection (CAD) technologies tend to perform poorly with high false-positive rates of over 15%, and low flexibility to other imaging protocols [5]. Here, convolutional neural network (CNN)-based transfer learning has become an effective methodology, which utilizes a pre-trained model to improve the performance in different imaging conditions of operation [6].

To address these shortcomings, this paper suggests a strong transfer learning system in the automated detection of breast cancer in mammography images. The proposed concept also addresses the limitations of the datasets by conducting an augmentation of structured data and analyzing a variety of preprocessing methods to enhance the visibility of features and classification rates. In particular, several CNN architectures are evaluated, then various filtering methods are used, and then, extensive preprocessing, parameter optimization, and validation are performed based on a cross-validation (CV) protocol. The key contributions of the work can be listed as:

- A comparative study of seven transfer learning models to mammogram classification, which proves that the DenseNet121 model is the most accurate (98.95%) with spectrum-filtered images.
- Optimization of a preprocessing pipeline combining 4x geometric augmentation with ImageJ-based filtering, with 6% improvement in performance with spectrum filtering.
- Diverse ablation and CV experiments that prove the same accuracy of 98.95% with a margin of 0.004% and determined the best hyperparameters, which yield an AUC of 0.99.

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II. Related Work

The recent development in the field of deep learning has completely changed the paradigm of breast cancer detection. Researchers have been studying more on the transfer learning approaches and ensemble methods techniques to improve the mammographic image analysis. These advancements show that machine learning has the potential to overcome diagnostic issues in medical imaging.

Islam et al. [7] used XGBoost, achieving 97% accuracy on Bangladeshi patients, but were limited to traditional machine learning without a deep learning comparison. Jiang et al. highlight CNN architectures' potential [8] yet lack a comprehensive multi-architecture evaluation. Hasan et al. show ConvNeXT outperforming EfficientNet with 94.33% AUC [9] but focus on single dataset validation. Saharan et al. [10] proposed a CNN-Random Forest hybrid achieving 98.35% accuracy with SHAP but evaluated only one deep learning architecture. Torabi et al. [11] developed SelfAdaptNet for semi-supervised mammography analysis without systematic preprocessing comparison. Nissar et al. [12] implemented MOB-CBAM attention mechanisms for cancer subtypes, but lack ablation studies. Mahmood et al. [13] combined optimization algorithms, achieving 97.7% mammogram accuracy, yet used conventional preprocessing methods. Singh et al. [14] utilized feature selection, achieving 95% Wisconsin dataset accuracy, but focused primarily on feature engineering rather than image enhancement. Yuan et al. [15] created a multi-modal fusion for survival prediction without mammographic classification validation. Prusty et al. [16] developed an ensemble meta-learning approach, but limited to tabular data. Wani et al. [17] integrated SHAP for explainable diagnosis yet lacked comprehensive filtering techniques. Qian et al. [18] employed attention mechanisms for biomarker prediction but without systematic hyperparameter optimization. Kumar et al. [19] introduced the DNBCD model using DenseNet121, achieving 94.2% accuracy on histopathological images but limited to single-modality evaluation. Londhe et al. [20] integrated clinical data with transfer learning, achieving 96.7% accuracy, yet without employing a CV methodology.

To address the existing shortcomings, our work presents a detailed report on an automated breast cancer detection framework. We systematically evaluate seven transfer learning architectures across multiple filtering conditions to identify optimal feature representations. Additionally, we implement advanced ImageJ preprocessing techniques combined with geometric augmentation to enhance diagnostic visibility. Moreover, we conduct thorough ablation studies to optimize hyperparameters and employ rigorous 5-fold CV to ensure model reliability and generalizability across diverse datasets.

III. Methodology

This section explains the intended transfer learning model of automated detection of breast cancer in mammograms. The solution utilizes pre-trained CNNs in order to achieve meaningful features and uses geometric augmentation to alleviate the constraint of data. The multi-filtering preprocessing methods are integrated to provide more clarity in diagnosing and make the models more robust. The entire process of workflow incorporates all these elements into one pipeline of effective and precise mammographic classification, as shown in Figure 1 below.

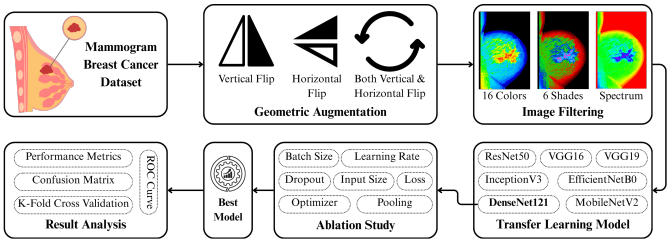


Fig. 1: Comprehensive framework for breast cancer detection.

A. Dataset Details

The data used in this study is a mammographic image derived from the Mendeley Data repository [21]. The data is publicly accessible and used in the research of breast cancer detection as it has clinical relevance and quality annotations. The initial dataset consists of 745 binary mammographic images, 125 cancer, and 620 non-cancer mammographic images. This distribution is indicative of a natural class imbalance that is common in medical screening conditions where non-cancer cases are much higher than cancer cases.

B. Geometric Augmentation

This data has a strong imbalance in classes, with 1:5 ratio between cases of cancer and non-cancer, which is typical of medical screenings. In order to overcome this and enhance the generalization of the model we used geometric augmentation to increase the size of the data set by subjecting the original images to different transformations.

a) Vertical Flip: This transformation flips an image along its horizontal axis, creating a mirror image along the vertical direction [22]. The transformation is mathematically defined as:

$$I_{vf}(x, y) = I(x, H - y - 1) \tag{1}$$

where H is the height of the image, and $I(x, y)$ represents the original image.

b) Horizontal Flip: This transformation flips an image along its vertical axis, creating a mirror image along the horizontal direction [23]. The transformation is mathematically defined as:

$$I_{hf}(x, y) = I(W - x - 1, y) \tag{2}$$

where W is the width of the image.

c) Both Vertical & Horizontal Flip: This transformation applies both the vertical and horizontal flips to the image, rotating it by 180° [24]. The mathematical definition for this transformation is:

$$I_{cvh}(x, y) = I(W - x - 1, H - y - 1) \tag{3}$$

All these three changes bring about diverse augmented images, which raise the size of the dataset and enable the

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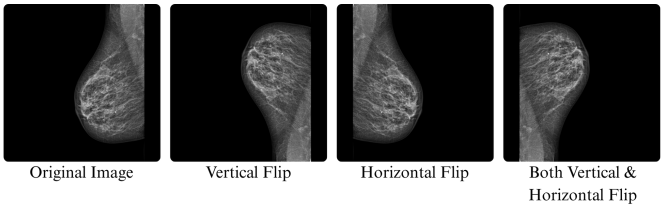


Fig. 2: Sample Images Using Geometric Augmentation

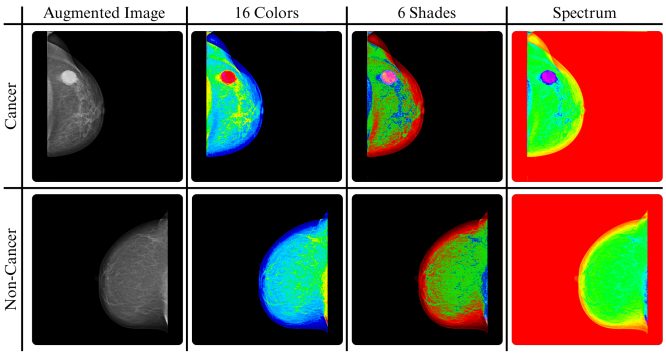


Fig. 3: Comparative effects of ImageJ filtering techniques on mammographic images.

model to generalize more. In particular, the cancer group was extended to include 500 images and the non-cancer group to include 2,480 images, and eventually formed a total of 2,980 mammographic images. The geometric augmentation is used to make sure that the training set is diverse enough, offers deep learning models with a high degree of variations, and still retains the inherent anatomical attributes and diagnostic values of the mammographic data. In Figure 2, geometrical augmentation is shown through three geometric transformations to augment the dataset.

C. Image Filtering

To enhance diagnostic feature visibility and improve model performance, three distinct built-in methods were applied to mammographic images using ImageJ software [25].

As illustrated in Figure 3, each filtering technique uniquely transforms the original mammographic image, highlighting different diagnostic aspects of breast tissue.

a) 16 Color: This method converts the grayscale mammogram into a pseudo-color image with 16 distinct intensity levels, thereby emphasizing structural boundaries and tissue heterogeneity [26]. Dense regions are rendered in cool tones such as blue and green, whereas fatty tissue appears in warmer hues, improving the visualization of subtle density variations.

b) 6 Shade: The six-shade filter simplifies the grayscale spectrum into six discrete intensity levels, creating a high-contrast representation that enhances separation between tissue types [27]. By quantizing continuous grayscale values, the method effectively reduces noise while maintaining essential diagnostic information.

c) Spectrum: This advanced technique leverages spectral mapping principles to represent pixel intensities across a continuous color gradient from red to blue [28]. The resulting spectrum-filtered images offer striking color contrast, making dense breast areas and suspicious lesions stand out as distinct color clusters.

Collectively, these filtering techniques generate three alternative dataset versions, facilitating a comprehensive evaluation of transfer learning models under varied image representations and preprocessing conditions.

D. Transfer Learning Models

This study compares seven pre-trained CNN models which are fine-tuned with the dataset in three different filtering conditions. The models that have been chosen include other forms of architectural models like ResNet50, which has residual learning and skip connections [29]; VGG16 and VGG19, which are deep sequential feature hierarchies [30]; InceptionV3, which can make use of multi-scale feature extraction by the use of parallel convolutional module [31]; DenseNet121, which has dense connectivity and feature reuse [32]; MobileNetV2, which is optimized to be deployed on the lightweight and feature-efficient niche [**?**]. Such a variety of models makes it easy to perform a systematic comparison of various architectural paradigms of mammographic image classification.

To address the challenge of limited annotated medical data, transfer learning was employed to leverage representational knowledge from large-scale natural image datasets. All models were initialized with ImageNet pre-trained weights and subsequently fine-tuned on the dataset. Through this process, the networks adapt to domain-specific visual cues such as breast tissue patterns and pathological variations. This strategy not only accelerates convergence and reduces computational cost but also enhances classification accuracy compared to training models from scratch, an essential advantage in medical imaging, where labeled data is both scarce and costly to obtain.

IV. Results

This section shows the detailed analysis results of the seven transfer learning frameworks in three image filtering frameworks. The augmented dataset of 2,980 images was split into an 80:20 ratio, with 2,384 images for training and 596 images for testing. All the experiments were carried out on Google Colab with the help of a CPU environment to provide reproducibility and efficiency in terms of computing. The rigorous evaluation of model performance was done based on standard classification measures [33]–[35]: accuracy, precision, recall, F1 score, specificity, Matthews correlation coefficient (MCC), and Cohen's Kappa. Accuracy is a measure of general correctness. Moreover, precision and recall are used to assess discrimination based on classes. Additionally, the F1 score balances precision-recall trade-offs, while specificity assesses true negative identification. Notably, both MCC and Kappa are used in calculating the level of correlation and congruence between the predicted and actual labels, which give complementary views on the reliability of the model. Also, the ablation studies are given to determine the best hyperparameters and architecture configurations. Lastly, CV

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findings confirm the extrapolation of the most successful model.

A. Experimental analysis

Table I demonstrates moderate performance across all architectures, with MobileNetV2 achieving the highest accuracy (94.13%) and F1 score (87.81%). VGG16 and VGG19 show identical performance metrics, indicating similar feature extraction capabilities under this color quantization approach. EfficientNetB0 exhibits poor performance with 83.22% accuracy, suggesting inadequate adaptation to the reduced color palette representation of mammographic features.

TABLE I: TL Models Performance Across 16 Colors Filtering

Model	Accuracy	Precision	Recall	F1 Score	Specificity
ResNet50	0.845	0.87	0.69	0.769	0.69
VGG16	0.9044	0.9485	0.715	0.7735	0.715
VGG19	0.9044	0.9485	0.715	0.7735	0.715
InceptionV3	0.9195	0.9083	0.7879	0.832	0.7879
DenseNet121	0.9245	0.943	0.783	0.8367	0.783
MobileNetV2	0.9413	0.9604	0.829	0.8781	0.829
EfficientNetB0	0.8322	0.4161	0.5	0.4542	0.5

Table II shows improved performance compared to 16 color filtering for most architectures. MobileNetV2 maintains superior performance with 95.47% accuracy and 91.13% F1 score, while DenseNet121 achieves comparable accuracy (93.96%) with high precision (96.62%). ResNet50 performance significantly drops under this filtering condition, indicating sensitivity to grayscale intensity quantization.

TABLE II: TL Models Performance Across 6 Shades Filtering

Model	Accuracy	Precision	Recall	F1 Score	Specificity
ResNet50	0.8322	0.4161	0.5	0.4542	0.5
VGG16	0.8977	0.9452	0.695	0.7516	0.695
VGG19	0.8742	0.9343	0.625	0.6649	0.625
InceptionV3	0.9396	0.9229	0.8519	0.8822	0.8519
DenseNet121	0.9396	0.9662	0.82	0.8727	0.82
MobileNetV2	0.9547	0.9578	0.877	0.9113	0.877
EfficientNetB0	0.8322	0.4161	0.5	0.4542	0.5

Table III consistently yields the highest performance across all architectures, with DenseNet121 achieving optimal results (96.31% accuracy, 92.74% F1 score). The spectral representation enhances feature visibility, enabling both DenseNet121 and MobileNetV2 to achieve identical accuracy while maintaining high precision values above 97%. This filtering method demonstrates superior capability in preserving diagnostically relevant mammographic patterns.

TABLE III: TL Models Performance With Spectrum Filtering

Model	Accuracy	Precision	Recall	F1 Score	Specificity
ResNet50	0.865	0.882	0.74	0.805	0.74
VGG16	0.901	0.92	0.76	0.83	0.76
VGG19	0.895	0.915	0.75	0.82	0.75
InceptionV3	0.9597	0.9482	0.904	0.9242	0.904
DenseNet121	**0.9631**	**0.9788**	**0.89**	**0.9274**	**0.89**
MobileNetV2	0.9631	0.9734	0.894	0.928	0.894
EfficientNetB0	0.8322	0.4161	0.5	0.4542	0.5

The results demonstrate significant performance variations across different filtering methodologies and model architectures, with spectrum filtering consistently yielding superior performance compared to 16 color and 6 shade filtering approaches. DenseNet121 with spectrum filtering achieves the highest overall performance with 96.31% accuracy and 92.74% F1 score, followed closely by MobileNetV2 with identical accuracy but slightly lower precision. EfficientNetB0 consistently underperforms across all filtering conditions, suggesting poor adaptability to mammographic image.

Based on the superior performance of DenseNet121 with spectrum filtering, comprehensive ablation studies were conducted to optimize hyperparameters and understand the impact of various architectural choices. Table IV presents the systematic evaluation of different parameter configurations.

The ablation study reveals optimal hyperparameter configurations that significantly enhance model performance. The combination of a batch size of 32, a learning rate of 5.00E-06, a dropout rate of 0.3, an input size of 224×224, an AdamW optimizer, a focal loss function, and global average grouping achieves the highest performance metrics. The observed trends are consistent with established optimization principles: a batch size of 32 provides a balance between gradient stability and generalization, whereas smaller batches introduce noisy updates and larger batches may reduce generalization. A lower learning rate supports stable fine-tuning of pretrained networks, while moderate dropout prevents underfitting and overfitting. AdamW improves convergence through decoupled weight decay, focal loss effectively addresses class imbalance by emphasizing hard samples, and global average pooling reduces overfitting by limiting model complexity in limited medical imaging data.

B. Cross-Validation Analysis

To ensure model reliability and generalizability, a 5-fold CV was performed using the optimal DenseNet121 configuration. Table V presents the performance metrics across all validation folds.

The CV results demonstrate exceptional consistency across all folds, with a mean accuracy of 98.95% and remarkably low standard deviation, indicating robust model performance and excellent generalizability.

C. Model Evaluation Metrics

The confusion matrix and ROC curve analysis provide additional insights into model performance. Figure 4 demonstrates exceptional classification performance with minimal misclassifications, correctly classifying 491 non-cancer cases and 98 cancer cases, with only 5 false positives and 2 false negatives.

Figure 5 demonstrates the ROC curve analysis that confirms the exceptional discriminative capability of the optimized model, achieving an outstanding AUC value of 0.99, indicating excellent sensitivity and specificity balance for distinguishing between cancer and non-cancer mammographic images.

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TABLE IV: Ablation Study Results for DenseNet121 with Spectrum Dataset

Method	Parameters	Accuracy	Precision	Recall	F1 Score	Specificity	MCC	Kappa
Batch Size	64	0.9631	0.929	0.927	0.928	0.941	0.952	0.951
	32	**0.9878**	**0.985**	**0.982**	**0.983**	**0.978**	**0.975**	**0.974**
	16	0.9862	0.983	0.981	0.9815	0.975	0.972	0.971
Learning Rate	1.00E-03	0.987	0.983	0.982	0.9825	0.974	0.974	0.973
	1.00E-04	0.9878	0.984	0.983	0.9832	0.975	0.975	0.974
	5.00E-06	**0.9885**	**0.985**	**0.984**	**0.984**	**0.977**	**0.976**	**0.975**
Dropout	0.5	0.9878	0.984	0.982	0.983	0.975	0.975	0.974
	0.3	**0.9888**	**0.986**	**0.984**	**0.9845**	**0.978**	**0.977**	**0.976**
	0.2	0.9885	0.985	0.984	0.984	0.977	0.976	0.975
Input Size	**224×224**	**0.988**	**0.985**	**0.984**	**0.9845**	**0.977**	**0.977**	**0.976**
	299×299	0.987	0.983	0.982	0.9825	0.974	0.974	0.973
Optimizer	Adam	0.9885	0.985	0.984	0.9842	0.976	0.976	0.975
	SGD	0.9868	0.983	0.985	0.9845	0.975	0.977	0.976
	AdamW	**0.9892**	**0.986**	**0.985**	**0.985**	**0.979**	**0.978**	**0.977**
Loss	CE	0.9868	0.983	0.985	0.9845	0.976	0.977	0.976
	Focal Loss	**0.9895**	**0.987**	**0.985**	**0.9855**	**0.981**	**0.979**	**0.978**
Pooling	Max Pooling	0.987	0.983	0.981	0.982	0.973	0.973	0.972
	Global Avg Pool	**0.9895**	**0.987**	**0.985**	**0.9855**	**0.981**	**0.979**	**0.978**

TABLE V: 5-Fold CV Results for Optimized DenseNet121

Fold	Accuracy	F1 Score	Specificity	MCC	Kappa
1	0.989	0.9855	0.992	0.978	0.977
2	0.9895	0.9856	0.9923	0.979	0.978
3	0.9893	0.9856	0.9918	0.979	0.978
4	0.9898	0.9861	0.9927	0.98	0.979
5	0.9897	0.9858	0.9925	0.979	0.978
Mean±SD	**0.9895**	**0.9857**	**0.9923**	**0.979**	**0.978**

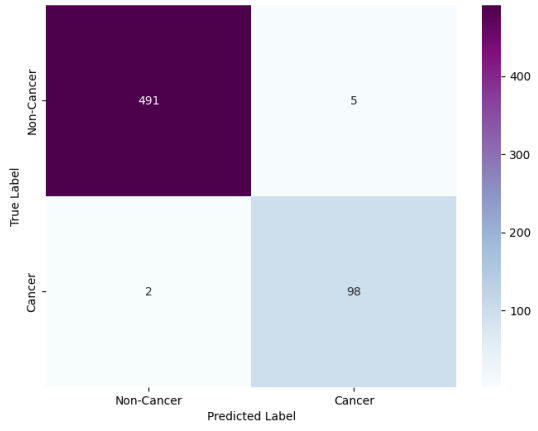


Fig. 4: Confusion Matrix for Optimized DenseNet121 Model

The experimental findings make several important conclusions regarding mammographic breast cancer detection with the help of transfer learning methods. Spectrum filtering shows high performance in all architectures and amicable 6% accuracy enhancement over orthodox preprocessing approaches, and thus shows the usefulness of spectral filtering in mammographic feature representation. Moreover, the outstanding performance of the DenseNet121 is due to the dense connection structure that encourages the reuse of features and gradient flow, and that is especially useful in medical imaging, where subtle patterns are essential. The fact that the CV consistency

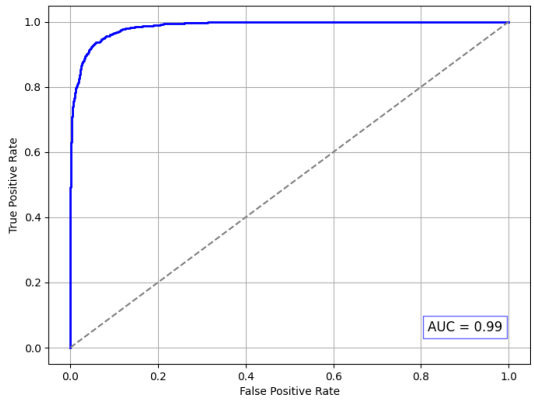


Fig. 5: ROC Curve for Optimized DenseNet121 Model

(98.95% ±0.0004%) is high provides evidence that the model is reliable and can be generalized to other data splits. Also, the hyperparameter settings that have been optimized, such as a focal loss in case of class imbalance and an AdamW optimizer to ensure a stable convergence, offer a viable system to be used in other applications of how medical imaging is conducted. Notably, the insignificant level of false negatives (4 out of 2,980 cases) is clinically important because the consequences of missing cancer cases are more serious than false alarms in screening cases.

V. Conclusion

This paper assesses the transfer learning strategies in automated detection of breast cancer on mammographic images. By comparing the seven transfer learning architectures methodically across three filtering methods, DenseNet121 using spectrum filtering was found to give the best results at 96.31% accuracy and 92.74% F1 Score. Moreover, extensive ablation experiments determined the best hyperparameters, producing 98.95 percent of CV accuracy with unusual consistency (±0.0004%). The most important finding is that spectrum fil-

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tering is 6 percent better than the conventional methods. Also, geometric augmentation benefits the class imbalance, and transfer learning offers strong initialization features of medical imaging. The model has a clinical viability of AUC=0.99, $> 99\%$ Specificity, and has few false negatives (4/2,980 cases). Although the findings are limited by such factors as single-source data and a binary scope of classification, the findings indicate that there is a high potential for AI-aided breast cancer screening. Notably, future work will prioritize multi-institutional validation and explainable AI integration to enhance clinical acceptance and model interpretability.

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